TEXORA · LECTURE SHEET

Process Control of Knitting

Knitting quality is decided by a short list of controlled variables — stitch length above all, then yarn input tension, take-down, machine condition and environment. Process control means measuring those variables on a schedule, against tolerances, and correcting the process before fabric goes out of specification.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 9 · LEVEL 3 TERM 1

1. The control hierarchy

PRIORITY	VARIABLE	MEASURED HOW	TYPICAL TOLERANCE*
1	Stitch length (per feeder/system)	Unrove one course, measure yarn length ÷ loops; or yarn-length counter per 100 revolutions	±2 % feeder to feeder
2	Yarn input tension	Tensiometer on running yarn; feeder calibration	Set per yarn; evenness matters most
3	Fabric dimensions after relaxation	CPC/WPC with glass/counter, width at several points	Per order spec (e.g. width ±2 %)
4	GSM	Cut-and-weigh (GSM cutter, 100 cm²) after conditioning	Commonly ±5 %
5	Machine condition	Needle/sinker inspection, cam and trick wear, oil and lint state	Zero broken/missing elements in action
6	Environment	Temperature/RH log in the shed	≈ 65 % RH working target for cotton

*Tolerances are factory-set examples, not universal standards — the order specification governs.

2. Stitch length control in depth

Stitch length is controlled at two levels. **Setting:** stitch-cam depth plus positive-feed delivery (yarn length per revolution divided by loops per revolution gives the nominal l). **Verification:** the unroving test — mark one course, extract it, measure its length under standard light tension, divide by the

number of loops (needles that knitted). Example (verified arithmetic): a course from a 2262-needle machine measures 11.31 m → l = 11,310 mm ÷ 2262 = **5.00 mm/loop**. Repeat per feeder (each feeder knits its own course) — a single feeder 3 % long prints a barré band every feeder-repeat. Modern frames log yarn consumption per feeder continuously; a drifting feeder is visible before fabric is.

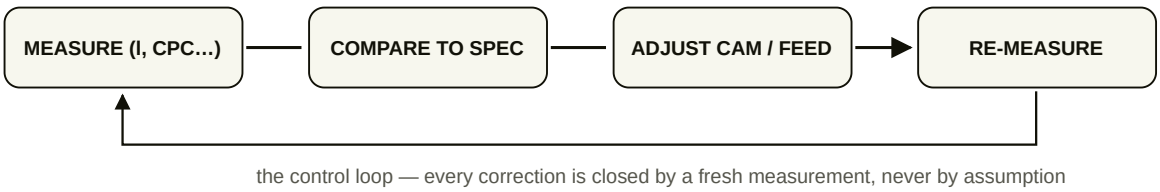


Figure 1 — Closed-loop control of stitch length. The same loop, with different measurements, controls width, GSM and course spacing.

3. Yarn and machine-side control

- **Yarn:** count variation, twist liveliness (spirality), hairiness and lot mixing are incoming controls — test per lot, segregate lots, and never let a new lot enter mid-piece. Wax/friction state affects input tension and hence effective loop length at the needle.
- **Needles and sinkers:** scheduled replacement by running hours and fault history; a single damaged needle is replaced at once (it prints every course). Lint removal and oiling on schedule — dry tricks raise friction, bend butts and vary stitch length around the circle.
- **Take-down:** uneven take-down (one side pulling harder) skews courses on the machine and distorts width; check roll hardness across the width.
- **Stop motions and detectors:** yarn-break, needle (hole) detectors and press-off protection tested each shift — a failed stop motion converts a 2-minute break into a destroyed piece.

4. Fault analysis — reading the fabric

FABRIC SIGN	DIRECTION	FIRST SUSPECTS
Barré bands	Course-wise, repeating with feeder cycle	One feeder's stitch length/tension; yarn lot change at one feeder
Needle line	Wale-wise, continuous	Damaged needle, dirty/bent trick at that wale
Holes / dropped stitches	Local, often one wale	Latch failure, clearing fault, yarn defect at that point

Thick/thin courses scattered	Random course-wise	Yarn count variation, tension spikes, knot passage
Width drift over the roll	Gradual	Feed calibration drift, temperature/RH change, take-down slip
Spirality after wash	Diagonal skew	Yarn twist direction/liveliness + unbalanced structure; finishing route

The diagnostic rule: **course-wise faults belong to feeders and yarn delivery; wale-wise faults belong to needles and tricks; once-per-revolution faults belong to rotating geometry (dial/cylinder); gradual drifts belong to calibration and environment.** This single rule routes most troubleshooting in Topics 4–7.

5. Worked control example

WORKED EXAMPLE — FEEDER CORRECTION FROM AN UNROVING TEST

Target l = 5.00 mm on the Topic 4 machine (2262 needles). Feeder 17's course measures 11.65 m → l = 11,650 ÷ 2262 = **5.15 mm**, i.e. 3.0 % long — outside the ±2 % band and enough to print barré.

Correction: reduce Feeder 17's positive-feed delivery by 3 % (or raise its stitch cam equivalently) — target course length = 2262 × 5.00 = 11.31 m. Re-knit, re-unrove Feeder 17, and confirm 11.31 m ± 0.23 m (±2 %) before releasing the roll.

Glossary

Unroving test — measuring stitch length by extracting and measuring one knitted course. **Conditioning** — bringing samples to standard atmosphere before weighing/measuring. **Press-off** — fabric coming off the needles after a yarn break, losing all loops of the affected courses. **Patrol** — scheduled in-run inspection round.

Key points

• Control stitch length first, per feeder, by unroving or yarn counters; close every adjustment with re-measurement.

• Read faults by direction: course-wise = feeders/yarn, wale-wise = needles, once-per-rev = geometry, drift = calibration/environment.

• Yarn lots, needle condition, take-down evenness and working stop motions are process control, not housekeeping.

Practice

1. A course unroved from a 2639-needle machine (30 in E28) measures 12.40 m. Find l, and state whether it passes a 4.70 mm ±2 % specification.
2. A roll shows one dark band every 72 courses on a 72-feeder machine. Localise the fault and name the first adjustment.
3. Why is GSM checked on conditioned, relaxed fabric rather than at the machine?
4. Give the control response when width drifts 1 % narrower over four hours with stitch length unchanged.

Answer key

1. l = 12,400 ÷ 2639 = **4.70 mm** (4.698) — exactly on target, passes (band = 4.61–4.79 mm).
2. One feeder of the 72 is delivering differently (its course repeats once per 72 courses): unrove that feeder's course, compare l to the target, recalibrate its positive feed/stitch cam, re-measure.
3. On the machine the fabric is under take-down tension and holds knitting-state dimensions; GSM by definition describes the fabric as delivered/used — relaxed and moisture-conditioned — otherwise tension and regain differences falsify the weight.
4. With l constant, suspect the relaxation/take-down side: check take-down tension and roll pull across the width, shed temperature/RH drift, and yarn lot change; verify CPC/WPC on relaxed samples before touching stitch cams (which would change l and GSM together).